

A Comprehensive Review of Biosignal Foundation Models

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Abstract—Foundation models are emerging as a promising approach to biosignal analysis, offering solutions to common challenges in learning from biosignal data, such as high variability, noise, or limited annotations. This survey provides a comprehensive review of the state-of-the-art in this rapidly advancing field. We introduce commonly used biosignals (including EEG, ECG, EMG, EOG, and PPG) and discuss their defining characteristics with respect to training large models. We present a structured survey of existing work across unimodal and multimodal settings covering data processing, feature extraction, model architectures, pre-training paradigms, and evaluation methods. We identify open challenges such as benchmarking, interpretability, and data availability and highlight possible directions for future research. By consolidating our findings from the current literature and opportunities for progress, we aim to guide the development of robust and universal foundation models for biosignals.

Index Terms—Foundation Models, Biosignals, Brain-Computer Interfacing (BCI), Electroencephalography (EEG), Electromyography (EMG), Electrooculography (EOG), Electrocardiography (ECG), Photoplethysmography (PPG)



1 INTRODUCTION

THE human body is constantly emitting biosignals: streams of measurable signals arising from the physiological processes occurring in our tissues, organs, and systems. These signals can be recorded from sensors placed inside or on the surface of the body, giving useful insights into what a person is doing and how their body is functioning. This ability to interpret meaningful information from biosignals has a broad societal impact. Clinically, biosignals are a primary diagnostic and monitoring tool. For example, electroencephalography (EEG) can be used to diagnose epilepsy [1] or sleep disorders [2], and electrocardiography (ECG) enables early detection of heart arrhythmia [3]. Beyond healthcare, biosignals are now being used in emerging applications such as brain-computer interfaces (BCI) [4].

The study of biosignals has a long history, beginning with visual inspection of raw signal data by trained clinicians. This method is still the preferred choice of many practitioners, but it can be unreliable, even with experienced experts, because much of the relevant information is not easily visible from an image or signal trace. Various signal processing methods have been used to extract latent information, including Fourier analysis and wavelet transform [5], [6]. Derived features would typically be coupled with statistical or rule-based models for a specific diagnostic task. With the advent of machine learning, classical models

such as Support Vector Machines (SVM) [7], random forests [8], and Hidden Markov Models (HMM) [9] have since been applied. These models often relied on using handcrafted features as input, not having the necessary complexity to effectively capture information from raw signals [10]. This issue started to be addressed by Deep Learning architectures, such as Convolutional Neural Networks (CNN) [11], Recurrent Neural Networks (RNN) [12] and its variants like Long Short-Term Memory (LSTM) [13]. While these approaches can show impressive performance when trained for specific purposes, they often struggle to generalise to samples acquired under different conditions from their training data. From a causal reasoning perspective, a biosignal recording is the end result of several variables; these include the relevant underlying physiological activity, but also the subject's intentions, their environment and any stimuli they may be presented with [14]. When the distributions of these variables shift, for example with an unseen subject or a new task, it can cause a deep learning model's performance to degrade sharply. Another major constraint is that they require supervised training with labelled data which may be limited.

In recent years, the rapid progress of foundation models has transformed the field of deep learning. These models are characterised by their ability to learn powerful representations of data which can generalise beyond the conditions of their pre-training data. Motivated by their successful use in computer vision and Large Language Models (LLM), foundation models are now being trained to model biosignal data. These models aim to overcome the limitations of earlier deep learning approaches; making use of large, diverse unlabelled data during pre-training, and promising better generalisation capabilities across downstream tasks.

In this survey we provide an overview of the emerging field of foundation models for biosignals, including single modality and multimodal models. **For the purpose of this**

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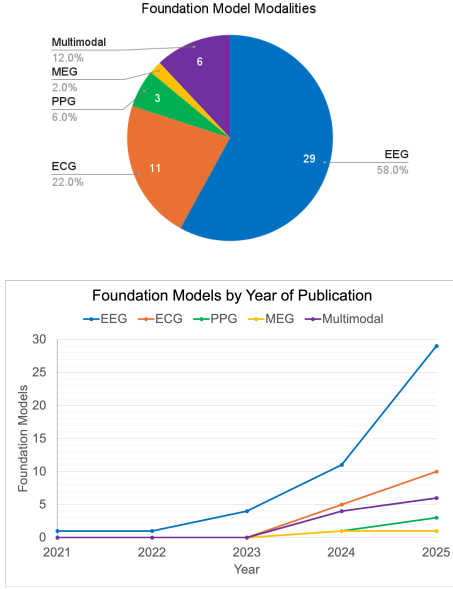


Fig. 1. Overview of current foundation model publications. Proportion of models by modality (top), and trends in cumulative number of publications by year (bottom).

study we define a **foundation model** as a model that has been pre-trained on large-scale data, and is designed to generalise to a range of tasks. Related articles have reviewed foundation models specific to the EEG modality [15], [16], and [17] covers applications of LLMs with EEG including "LLM-style models". Other surveys have reviewed models across multiple biosignal domains such as [18] which includes Transformers, and [19] which covers multiple deep generative model architectures. However, these do not include foundation models according to our definition. While [20] has also provided a survey of biosignal foundation models from a conceptual perspective, we present a detailed analysis of architectural and pre-training design choices including fusion techniques used for alignment between multiple biosignal modalities.

The rest of the paper is organised as follows. Section 2 outlines our search strategy and gives an overview of the literature. Section 3 provides the necessary background context on biosignals and the foundation model paradigm. Sections 4, 5, 6 and 7 present the main results of our survey in which we systematically review current biosignal foundation models, categorised by modality. Finally, in Section 9 we discuss open challenges and promising future directions for research. A glossary of acronyms used within this article is provided in Table 1 for reference.

2 SEARCH STRATEGY

For this survey we searched for relevant articles across several platforms including: arXiv, IEEE Xplore, DBLP and PubMed. We performed a search for each of the following acronyms: "EEG", "EMG", "EOG", "ECG", "EKG", "PPG" (and "biosignal") in conjunction with "Foundation Model" or "Pre-trained". In this paper, we include articles published or archived before 26th August 2025. Combined across platforms and searches, these produced a total of 192 results.

We then removed the duplicates and scanned the content of each paper to determine whether the authors develop a foundation model according to our definition as given above. We specifically exclude papers in which a pre-trained model from another publication has been used directly, and the authors do not pre-train any additional biosignal encoding modules. A total of 48 articles remain after this screening process.

We found that the majority of proposed models were trained exclusively on EEG data, while a few unimodal models for ECG and PPG have also been published within the past two years. It should be noted that two of the articles presented separate unimodal models for different modalities; these include [21] for ECG and PPG, and BIOT [22] for EEG and ECG. No foundation models were found specifically for EMG or EOG signals, however these are included in several multimodal models. One unimodal model was also found for Magnetoencephalography (MEG) although this modality was not explicitly referenced in the search terms. A full breakdown of the models by modality and year of publication is given in Fig. 1. From the cumulative plot it can be seen that the earliest biosignal foundation models were developed for EEG, however the number of models for this modality only started to grow rapidly from 2023 onward. This could be due to the promising capabilities shown by open source models published that year, including BIOT [22], BrainBERT [23] and Brant [24]. Similarly, following their success in brainwave modelling, several foundation models for ECG and multimodal biosignal data have also emerged since 2024.

3 BACKGROUND

3.1 Biosignals

Biosignals are measurements that capture the physiological activity of a living organism over time. Examples of common modalities are illustrated in Fig. 2, including EEG for monitoring brain activity, ECG for heart function, electromyography (EMG) for muscle activity, electrooculography (EOG) for eye movement, and photoplethysmography (PPG) for changes in blood volume. Biosignals are usually recorded as time series data and reflect complex biological interactions across multiple temporal and frequency scales. A single modality may also be recorded from multiple channels and therefore contain spatial information based on the absolute and relative placements of the sensors in/on the body.

Biosignal analysis faces several common challenges due to the unique characteristics of physiological data compared to other modalities such as text or images. Many recorded signals are weak and easily contaminated by noise from various sources (movement, electrical interference etc.) leading to a low Signal-to-Noise ratio (SNR). Differences in recording device hardware, sensor placement, and recording conditions can introduce further inconsistencies in the statistical properties of the data. Inter-subject variability also poses the challenge of handling the variance caused by anatomical or physiological differences between individuals.

AFIB	Atrial Fibrillation	InfoNCE	Information Noise Contrastive Error
AUPRC	Area Under Precision-Recall Curve	LLM	Large Language Model
AUROC	Area Under Receiver Operating Characteristic Curve	LSTM	Long Short-Term Memory
BCI	Brain Computer Interface	LVH	Left Ventricle Hypertrophy
BPE	Byte-Pair Encoding	MAE	Masked Autoencoder
CLS	Classification	MEG	Magnetoencephalography
CNN	Convolutional Neural Network	MLP	Multi-layer Perceptron
CPE	Conditional Positional Encoding	MSE	Mean Squared Error
DFT	Discrete Fourier Transform	NLP	Natural Language Processing
DPSS	Discrete Prolate Spheroidal Sequences	PPG	Photoplethysmography
ECG	Electrocardiography	PSD	Power Spectral Density
EEG	Electroencephalography	PSG	Polysomnography
EKG	Electrocardiography	PU	Positive Unlabelled
EMA	Exponential Moving Average	PVC	Premature Ventricular Contraction
EMG	Electromyography	REM	Rapid Eye Movement
EOG	Electrooculography	rFFT	Real-valued Fast Fourier Transform
ERN	Error Related Negativity	RNN	Recurrent Neural Network
ERP	Event Related Potential	RoPE	Rotary Position Encoding
FC	Fully Connected	RVQ	Residual Vector Quantisation
FFT	Fast Fourier Transform	SNR	Signal-to-Noise Ratio
fMRI	Functional Magnetic Resonance Imaging	SSL	Self Supervised Learning
GAT	Graph Attention Network	SSM	State-Space Model
GNN	Graph Neural Network	SVM	Support Vector Machine
HMM	Hidden Markov Model	VQ	Vector Quantisation
iEEG	Intracranial EEG		

TABLE 1
Glossary of acronyms.

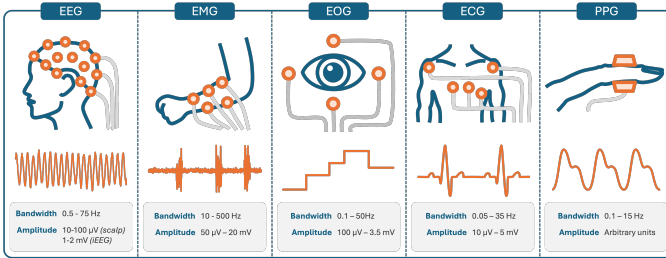


Fig. 2. Illustration of five common biosignals (EEG, EMG, EOG, ECG and PPG) with approximate bandwidths and amplitudes.

3.1.1 Electroencephalography (EEG)

EEG records small voltage fluctuations generated by the electrical activity of neurons in the brain. The signals primarily originate from the potentials of pyramidal neurons in the cerebral cortex. When many neurons fire synchronously, the resulting currents create measurable electrical fields that can propagate through the brain tissue, cerebrospinal fluid, skull and scalp. To record EEG, electrodes are either placed on the scalp or surgically implanted in the brain (known as intracranial EEG or iEEG).

EEG has a relatively high temporal resolution compared to other brain activity monitoring systems like functional magnetic resonance imaging (fMRI), on a scale of milliseconds vs. seconds. It is therefore able to capture fast dynamics such as the onset of seizures, or changes in attention level. In contrast, the spatial resolution is lower; scalp EEG recording

devices typically arrange sensors according to the standard international 10-20 system of 21 electrodes, and even the densest montages contain ~ 200 electrodes compared to sub-millimeter precision achieved in High-Resolution fMRI. Analysis of scalp EEG can also be challenging due to its low SNR compared to iEEG, although it has the advantage of being non-invasive and more cost-effective, thus enabling the collection of large-scale datasets.

3.1.2 Electromyography (EMG)

EMG measures the electrical activity generated by muscle fibres through electrodes placed on the skin or inserted into muscle tissue. These signals can be used in diagnosis to detect neuromuscular disorders [25], or to control prosthetics by decoding muscle signals and translating them into commands [26]. The waveforms of EMG are highly stochastic and irregular, reflecting spikes in motor unit action potentials during muscle contractions. EMG signals are naturally highly dependent on sensor placement and can vary greatly between individuals due to anatomical differences.

3.1.3 Electrooculography (EOG)

EOG tracks eye movement using bipolar electrodes that measure the potential difference between two points near the eye. The eye acts as an electrical dipole due to the potential difference between the retina and cornea, generating a strong electrical signal when it moves or blinks. This can be used to monitor visual attention, detect certain

sleep stages such as REM [27], or control interfaces via eye gestures [28]. However, the strength of the electrical signal can be a hindrance when recording other modalities: "ocular artifacts" occur when EOG signals are picked up by nearby EEG channels, which can be difficult to disentangle from true neural activity.

3.1.4 Electrocardiography (ECG)

ECG (also known as EKG) is one of the most widely used biosignals in medicine, capturing the rhythms of the heart. This is achieved by recording the potential differences caused by depolarisation and repolarisation of the heart muscle. In clinical settings, a standard 12-lead ECG is used to record heart activity from electrodes placed on the chest and limbs. The multi-channel setup provides detailed information on how electrical impulses travel through the heart from multiple perspectives, enabling localisation of abnormalities and pathologies. Single-lead or wearable ECG devices are becoming increasingly common for everyday continuous heart monitoring.

3.1.5 Photoplethysmography (PPG)

PPG measures blood volume changes within arteries via a small light sensor, often placed on the finger or wrist. It is commonly used in many wearable devices to track heart rate, blood oxygen or estimated stress level. PPG is convenient and inexpensive to record for long periods of time, but it is sensitive to motion artifacts and environmental conditions.

3.2 Foundation Models

This section provides the necessary context regarding: the definition of foundation models, the most popular model architectures, the data used for training foundation models, how such models are trained from unlabelled data, and the strategies used for downstream evaluation.

3.2.1 Definition of Foundation Models

A foundation model is a large model which is pre-trained on diverse data to learn information-rich representations that can be used for a range of downstream tasks [29]. Unlike traditional task-specific deep learning models, a foundation model aims to capture universal representations across domains and/or modalities, serving as a general-purpose backbone model.

Formally, let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ denote a large pretraining dataset where x_i may correspond to input signals from various modalities and/or tasks and y_i to task-related labels. A foundation model learns a function:

$$f_\theta : \mathcal{X} \rightarrow \mathcal{Z} \quad (1)$$

where $\mathcal{Z}$ is a shared latent space and θ are trainable parameters. The pretraining objective of a foundation model can be generally formulated as

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{x \sim \mathcal{D}} [\mathcal{L}(f_\theta(x), t(x))], \quad (2)$$

where $t(x)$ defines a self-supervised target (e.g., masked prediction, contrastive learning prediction, or next-step prediction). After pretraining, f_θ can be adapted to specific downstream tasks through task-specific model fine-tuning.

3.2.2 Model Architectures

Each foundation model is composed of a main backbone, preceded/proceeded by smaller modules. For example, fully connected (FC) layers or convolutions may be used for preliminary encoding before passing to the backbone, or they could be used for further decoding of the backbone output. The most common backbones for foundation models are Attention-based (e.g. Transformers [30]) and State-Space Model (SSM)-based (e.g. Mamba [31]).

Transformers were introduced for sequence-to-sequence learning tasks through improved handling of long-range dependencies. Its core mechanism is self-attention [32], enabling the model to weigh relationships between all elements, or "tokens", of the input sequence in parallel rather than sequentially processing tokens. Given a sequence of tokens $X = [x_1, \dots, x_T] \in \mathbb{R}^{T \times d}$, each transformer layer computes representations as:

$$\begin{aligned} H &= \text{Attention}(XQ, XK, XV) \\ &= \text{softmax}\left(\frac{XQ(XK)^\top}{\sqrt{d_k}}\right)XV, \end{aligned} \quad (3)$$

where Q, K, V are learnable projection matrices and d_k is the key dimensionality. For each token x_t , its representation h_t is updated as a weighted sum of all token values, enabling the model to capture long-range dependencies.

The Transformer is built from stacked encoder and decoder blocks, wherein each encoder block contains multi-head self-attention modules and position-wise feedforward networks, and the decoder blocks additionally incorporate cross-attention to the outputs of the encoder. The original formulation also adds sinusoidal positional encodings to the input embeddings to capture information about the order of tokens in the sequence. Foundation models can use the full encoder-decoder structure, although many approaches choose to use only the encoder blocks (BERT [33] style) or only the decoder blocks (GPT [34] style).

Transformer-based foundation models instantiate the general function f_θ (the backbone of the foundation model) through stacked self-attention and feed-forward layers that operate on input token embeddings. While Transformers provide global context modelling, their $\mathcal{O}(T^2)$ attention complexity limits scalability for long signals.

State-Space Models constitute an alternative to transformers. These architectures model temporal dynamics using continuous-time latent states governed by a linear dynamical system. Given a sequence of tokens $X = [x_1, \dots, x_T] \in \mathbb{R}^{T \times d}$, an SSM implements f_θ via parameterised state transitions and can be expressed as a continuous time recurrence:

$$h'(t) = Ah(t) + Bx(t), \quad y(t) = Ch(t) \quad (4)$$

where $x(t)$ is the input, $h(t)$ is the hidden state, and $y(t)$ is the output at time t . A , B , and C are learnable parameters. In practice, the system is discretised to process sequences of tokens:

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t, \quad y_t = Ch_t \quad (5)$$

where $\bar{A} = \exp(e^{\Delta A})$ and $\bar{B} = [\exp(e^{\Delta A}) - I]A^{-1}B$. The Mamba architecture, a recent advancement in state-space

modelling, introduces *selective scanning* in which $\bar{A}$ and $\bar{B}$ are modulated by linear projections of the current token:

$$h_t = (g_t^A \odot \bar{A})h_{t-1} + (g_t^B \odot \bar{B})x_t, \quad y_t = Ch_t \quad (6)$$

$$g_t^A = \sigma(W_A x_t), \quad g_t^B = \sigma(W_B x_t) \quad (7)$$

where W_A and W_B are learnable weight matrices.

This allows the transition dynamics to emphasise relevant inputs and suppress irrelevant ones. Unlike self-attention which computes pairwise similarities across all tokens, this mechanism scales linearly with sequence length $\mathcal{O}(T)$. Mamba-2 [35] extends Mamba by learning $\bar{A}$ and $\bar{B}$ directly, without the need for the continuous-to-discrete mapping with double exponentials, thereby improving training stability for long sequences.

Both Transformers and SSMs implement the same abstract mapping:

$$f_\theta : \mathcal{X} \rightarrow \mathcal{Z}, \quad h_t = f_\theta(X), \quad (8)$$

However, they differ in how temporal dependencies are represented: Transformers use content-based global interactions while Mamba uses learned state transitions. These choices reflect different inductive biases in f_θ . Transformers emphasise flexible context aggregation, whereas Mamba emphasises efficient continuous-time dynamics modelling.

3.2.3 Self-Supervised Training Objectives

Traditional deep learning models are trained with supervised learning on labelled data for each downstream task. In contrast, foundation models rely on one or more types of Self-Supervised Learning (SSL) to learn general-purpose representations from large amounts of unlabelled data. Mathematically, all SSL strategies can be expressed as instances of the generic foundation model objective (eq. 2), where the choice of self-supervised target function $t(x)$ and loss $\mathcal{L}$ defines the particular training paradigm. Popular SSL strategies include masked autoencoding (MAE), autoregressive training, and contrastive learning.

In **masked autoencoding**, masking is applied either directly to input data or in the latent space, and the model is trained to reconstruct masked portions. Mathematically, the target $t(x)$ is defined as the masked portion of the input itself, i.e. $t(x) = x_\Omega$ for a subset of masked positions Ω . The model is trained to reconstruct the missing components from the visible ones:

$$\hat{x} = f_\theta(x_{\setminus\Omega}), \quad \mathcal{L} = \frac{1}{|\Omega|} \sum_{i \in \Omega} \|\hat{x}_i - x_i\|. \quad (9)$$

Here, $\hat{x}_i$ denotes the reconstructed value for token x_i , and the loss measures reconstruction fidelity over the masked positions.

Autoregressive models define the target as the next token or segment in the sequence, $t(x) = x_t$ given all previous tokens $x_{<t}$. Using cross-entropy loss, the model is trained to learn the conditional distribution $p_\theta(x_t | x_{<t})$:

$$\mathcal{L}(\theta) = - \sum_{t=1}^T \log p_\theta(x_t | x_{<t}). \quad (10)$$

This training method is used commonly by many Transformer decoder-based architectures (e.g., GPT-style models).

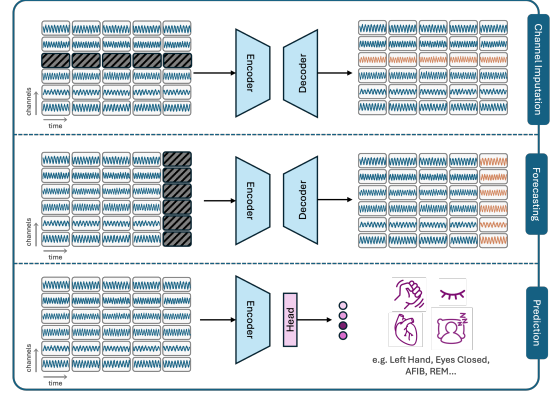


Fig. 3. Evaluation strategies in biosignal foundation models including generative and discriminative downstream tasks.

Contrastive learning methods define the target $t(x)$ implicitly through similarity relationships between positive and negative pairs in the latent space $\mathcal{Z}$. Contrastive learning encourages the model to minimise the distance between "positive pairs" while maximising distance between "negative pairs". The SimCLR [36] framework defines positive pairs as augmented views of the same input. In the CLIP [37] approach, positive pairings are predefined between samples from different modalities.

Let $z_i = f_\theta(x_i)$ denote the embedding of sample x_i . For a positive pair (i, j) and a set of negatives $\neq j$, the InfoNCE loss is defined as:

$$\mathcal{L} = - \log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^N \exp(\text{sim}(z_i, z_k)/\tau)} \quad (11)$$

where $\text{sim}(\cdot, \cdot)$ denotes a similarity metric (e.g., cosine similarity) and τ is a temperature parameter. Here, $t(x)$ is implicit: it defines the "correct" pairing that should yield maximal similarity in the learned latent space.

3.2.4 Evaluation Methods

Pre-trained foundation models are evaluated by assessing their performance on various downstream tasks including classification, regression, forecasting and imputation (Fig. 3 illustrates examples for biosignal foundation models that we examine in this work). To gauge performance on discriminative tasks, a lightweight prediction head is usually added as a final layer to the model. The pre-trained model can then be adapted with full fine-tuning, meaning all weights are further trained with downstream data, to estimate the performance in real-world settings. Linear probing is often applied, where the foundation model is frozen and only the classification head is trainable: this provides insight on the efficacy of pre-training and the model's true capability to generalise. For forecasting or imputation tasks, zero-shot learning is also applicable, where the pre-trained model is used directly on unseen data with no additional module training. Commonly used evaluation metrics include accuracy, balanced accuracy, AUPRC, AUROC, F1 and MSE.

3.3 Biosignal Inputs to Foundation Models

To learn effective representations, foundation models require large datasets for pre-training. For biosignals this has only become feasible in recent years as more publicly available datasets have been released. Diversity in this pre-training data is essential for improved generalisation capability [38], which in the context of biosignals means including recordings from multiple devices, subjects, sessions and paradigms (i.e. the context under which the signals are collected).

The diversity of data then poses a challenge in aligning samples with different properties. For example, recordings are typically resampled to the same frequency during offline pre-processing to handle data acquired from multiple devices operating at different sampling rates. Biosignals are often multi-channel, where each channel corresponds to a different sensor location: it may be necessary for models to capture this spatial information whilst also managing heterogeneity in electrode placement between recordings. The inclusion of multiple biosignals introduces the additional challenge of aligning across modalities, which has been approached in various ways by multimodal foundation models (see Section 7).

Some biosignal models take raw signals directly as input, represented as $\mathbf{x} \in \mathbb{R}^{C \times T}$ with C channels and T time steps. However, many approaches take inspiration from time series models like PatchTST [39] and segment the sequence into short patches, i.e. $\mathbf{p} \in \mathbb{R}^{C \times N \times P}$. In many cases, only raw time series data are used, although samples can be augmented with manually extracted features such as Power Spectral Density (PSD) [40].

4 EEG FOUNDATION MODELS

In this section we review 29 EEG foundation models, spanning a range of architectural designs. The models vary in number of trainable parameters, in datasets used for pre-training, and in the scale of this data. One might expect some correlation between these values, however in Fig. 4 we see no clear relationship between the two. It should be noted that this visualisation does not provide a complete picture due to the exclusion of publications that do not explicitly report these sizes.

4.1 EEG Data and Features

In recent years, a large number of publicly available EEG datasets have been released, collected under various paradigms. These include motor execution/imagery, event related potentials (ERP) in response to stimuli, sleep staging, emotion recognition, and clinical contexts such as seizure detection and mental disorder diagnosis. Table 7 gives an overview of commonly used datasets in training and evaluation of EEG foundation models. In particular, we found that the Temple University Hospital archive of clinical recordings (TUEG [41] and subsets thereof) is frequently used due to its large corpus of 25,000+ recordings from over 10,000 subjects.

When EEG is used as input data to a foundation model, sequences of raw signals can be used, as in EEGM2 [42] and EEGTrans [43]. The most common practice is to segment

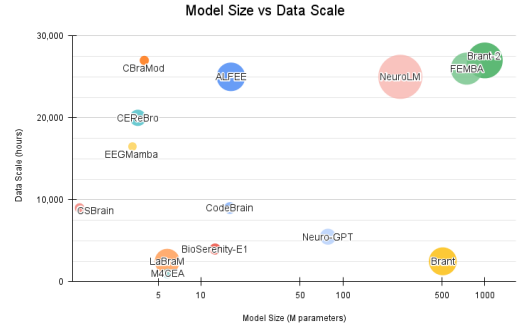


Fig. 4. EEG foundation model size vs pre-training data scale. Where a range of model sizes are given, the minimum is plotted on the X-axis while bubble area is proportional to maximum size. While no clear relationship is present, it can be seen that the largest models have been pre-trained on the most data.

the signal into short non-overlapping patches, although overlapping patches are used in BIOT [22]. Based on domain knowledge of EEG, the signals may be split into frequency bands: δ (0.5-4 Hz), θ (4-8 Hz), α (8-13 Hz), β (13-30 Hz), and γ (>30 Hz), where each band has a known association with different brain states. In practice, the exact frequency cutoffs between bands can differ, and γ is sometimes further divided into multiple bands [24], [44]. A common step in offline pre-processing is to perform highpass/lowpass filters in order to remove noise at low frequencies (such as from blinking) and high frequencies (such as muscle artifacts). For example, NeuroGPT [45] performs a bandpass filter at 0.5-100 Hz for all samples, and EEGPT [46] performs lowpass filters at different frequencies depending on the dataset.

Manually extracted features can also be passed directly into the model instead of the raw signal, as in MENDR [47] where wavelets from each frequency band are used as input. Alternatively, frequency features can be extracted and combined with the time series input; this is achieved in CBrMod [48] where the Fast Fourier Transform (FFT) energy vector is encoded separately, and added to the temporal encoding before entering the shared backbone.

In addition to time and frequency, multi-channel EEG contains spatial information from electrode placement, and relationships between the activity from different sensors' respective brain regions. In some cases spatial information is ignored completely, essentially proposing a single-channel model, as in BrainBERT [23]. Other models use a fixed number and order of channels for input data, thereby tying certain features to specific electrodes [45], [49], [50], although this method limits flexibility when pre-training samples are acquired from different sets of electrodes. Instead, many approaches handle channel heterogeneity by incorporating spatial information with embedding vectors.

4.2 Spatial and Temporal Embeddings

Embeddings contain spatial or temporal information about a sample, and are typically combined with signal features by addition or concatenation. They are incorporated before entering the backbone of the model, sometimes directly combined with input data patches [51], [52].

Spatial embeddings inform the model about which electrode the sample is from. This usually takes the form of a learnable parameter: given a global list of all electrodes included in the pre-training data $\mathbf{C} = \{c_1, \dots, c_G\}$, a parameter $\mathbf{se} \in \mathbb{R}^{G \times d}$ of the same length G , and embedding dimension d can be learned. A sample x from channel c_i would then be combined with the corresponding embedding $\mathbf{se}_i$. In contrast, Bioserenity-E1 [53] and LEAD [54] use fixed channel-wise embeddings, made possible by the predetermined channel number and order in the input data.

Following convention, temporal embeddings are added once the input has been patched and/or tokenized to inform the model of the temporal order of patches/tokens relative to the whole sequence. These embeddings are commonly implemented as a learnable parameter, as in Brant [24], CEReBrO [55], EEGFormer [51] among others. Another popular approach is to apply sinusoidal positional encodings [43], [53], [56], as proposed in the original Transformer [30]. An alternative to sinusoidal embeddings is Rotary Position Embedding (RoPE) [57] which applies rotation by an angle proportional to the token's position: this is applied in ALFEE [58], DIVER-0 [59] and EEGPT [46].

EEGMamba [60], inspired by work in Vision Transformers, uses conditional positional encoding (CPE) [61] where a depthwise convolution is applied to inject local positional bias. CBraMod [48] extends this by proposing an asymmetrical version of CPE to encode both spatial and temporal information. This consists of a depthwise 2D convolution layer with an asymmetrical kernel, designed to capture long-range spatial information and short-range temporal information. This approach was also later adopted in MENDR [47].

4.3 Model Architectures

All EEG foundation models included in this survey consist of Transformer or Mamba backbones, combined with additional lightweight modules. Table 2 provides a brief description of architecture and pre-training objectives for each model. Links to source code for all open-source models are also provided in Appendix C.

Preliminary Encoding and Convolutions

Convolutions over the time dimension are commonly applied as a preliminary encoding step before the backbone [58], [59], [68]. In some models, such as CSBrain [64] and EEGM2 [42], convolutions are applied at multiple temporal scales and then aggregated. Brant-2 [62] uses two CNNs to model time and frequency characteristics separately, and the concatenated features are inputted to the Transformer. Channel mixing can occur later, in a separate spatial CNN, as in CSBrain [64], or in spatio-temporal convolutions, as in FEMBA [65]. Convolutions may also be used after the backbone, as a decoder; this is the case in EEGTrans [43] and SS-EMERGE [56]. In some architectures such as BIOT [22], the initial encoding is achieved through simple FC layers.

Transformers

Transformers were found to be the most common backbone of EEG foundation models, in fact, the use of multiple Transformers throughout the pipeline is not uncommon. Many choose a BERT-style encoder-only version, such

as ALFEE [58], BENDR [49] and BrainBERT [23]. Others opt for a GPT-style decoder-only Transformer, such as BrainGPT [50], EEGPT [46] and Neuro-GPT [45]. An exception is EEGTrans [43] which uses a full encoder-decoder Transformer as its backbone.

Some models, including ALFEE [58], Brant [24], and LEAD [54], implement separate spatial and temporal Transformers, each consisting of multiple stacked blocks. Others, like CEReBrO [55], use alternating spatial and temporal blocks within one Transformer. CBraMod [48] proposes a combined spatio-temporal Transformer block in which standard multi-head self-attention is replaced by "Criss-Cross Attention": input embeddings are projected to multiple heads and each head applies attention across either channels or time.

Considering the spatial structure and inter-channel relationships present in EEG data, it can be useful to model this as a graph. This is achieved in GEFM [66] (an extension of BENDR [49]) by constructing a graph structure where each node represents an electrode, and edge weights are dependent on the physical distance between pairs of sensors. This fixed graph is used by a Graph Neural Network (GNN) as an initial step before encodings are passed to the BENDR model (a wav2vec 2.0 CNN [70] followed by a Transformer encoder). With a similar spatial graph, MENDR [47] takes inspiration from Graph Attention Networks (GAT) [71] and uses modified Transformer encoder blocks with GAT as the attention mechanism. SS-EMERGE [56] utilises separate Time Oriented and Feature Oriented GAT Transformers, where the graph nodes represent time points and latent features respectively.

Vector Quantisation

In some cases the foundation model is pre-trained in two stages, these include: BioSerenity-E1 [53], CodeBrain [63], EEGTrans [43], LaBraM [52] and NeuroLM [69]. In the first stage, a tokenizer is trained to generate meaningful tokens from input samples, the pre-trained tokenizer is then used in training a second stage foundation model. Contrary to the other two-stage models, the second stage of NeuroLM [69] does not involve training a new model, and instead adapts a pre-trained LLM by expanding its vocabulary to include EEG tokens (learned during the first stage).

In all two-stage EEG models we found that the tokenizer contains a vector quantisation (VQ) Codebook [72]. After input has been encoded to some latent representation h , a learnable codebook $\mathcal{V} = \{v_1, \dots, v_K\} \in \mathbb{R}^{K \times D}$ is used to lookup the nearest code to h :

$$z_i = \underset{j}{\operatorname{argmin}} \|h_i - v_j\| \quad (12)$$

In some cases, more than one codebook is learned, such as CodeBrain [63] which has separate codebooks for time and frequency features. EEGTrans [43] uses a Residual VQ (RVQ) [73] composed of a series of codebooks. Given the input h , the nearest code v is found in the first codebook, then the nearest code to the residual (i.e. the distance between h and v) is found in the next codebook, and this process is repeated through all the codebooks before aggregating the results. It should be noted that EEGFormer [51] also uses a VQ codebook, but this is trained in one stage simultaneously with the main foundation model.

Model	Architecture	SSL Objective	Model Size	Open source
ALFEE [58]	Temporal CNN, Spatial Transformer, Temporal Transformer, Decoder Transformer	Masked patch reconstruction, Signal forecasting	16.3M-540M	No
BENDR [49]	wav2vec CNN, Transformer	Contrastive learning	33M	Yes
BioSerenity-E1 [53]	1st stage: Temporal CNN, VQ Codebook, Decoder; 2nd stage: Temporal CNN, Transformer	1st stage: DPSS reconstruction, Codebook loss; 2nd stage: Masked code prediction	12.6M	No
BIOT [22]	FC Tokenizer, Transformer	Masked embedding reconstruction	3.2M	Yes
BrainBERT [23]	Transformer	Masked STFT/superlet reconstruction	43.2M	Yes
BrainGPT [50]	Temporal Transformer with causal attention	Autoregressive next patch prediction	1.5M-1.1B	No
Brant [24]	Temporal Transformer, Spatial Transformer	Masked patch reconstruction	505.7M	No
Brant-2 [62]	Temporal and Frequency CNNs, Temporal Transformer, Spatial Transformer	Masked patch reconstruction, Forecasting	1B	No
CBraMod [48]	Spatio-Temporal Transformer with "Criss-Cross Attention"	Masked patch reconstruction	4M	Yes
CEReBrO [55]	Alternating Spatial and Temporal Transformer	Masked token reconstruction	3.6M-85M	No
CodeBrain [63]	1st stage: CNN, Transformer, Time and Frequency VQ Codebooks, Time and Frequency Transformers; 2nd stage: Temporal CNN, Convolutional SSM, Sliding Window Attention	1st stage: Reconstruction (patch, FFT amplitude and phase), Codebook loss; 2nd stage: Masked code prediction	16M	No
CSBrain [64]	Multi-scale Temporal CNN, Multi-scale Spatial CNN, Spatio-Temporal Transformer	Masked patch reconstruction	1.4M-4.9M	No
DIVER-0 [59]	Temporal CNN, Transformer	Masked patch reconstruction	-	No
EEGFormer [51]	Temporal Transformer, VQ Codebook, Transformer	Masked patch reconstruction, Codebook loss	-	No
EEGM2 [42]	Multi-scale Temporal CNN, Encoder Mamba-2, Mediator Mamba-2, Decoder Mamba-2	Signal reconstruction, rFFT reconstruction	0.3M-4.5M	Yes
EEGMamba [60]	Temporal CNN, Alternating forward/backward Mamba-2 blocks	Masked patch reconstruction	3.3M	Yes
EEGPT [46]	Encoder Transformer, Predictor Transformer, Reconstructor Transformer; Momentum Encoder Transformer	Masked patch reconstruction	4.7M-25M	Yes
EEGTrans [43]	1st stage: Temporal CNN, RVQ Codebook, CNN Decoder; 2nd stage: Transformer, CNN Decoder	1st stage: Reconstruction (signal, mean and std), Codebook loss; 2nd stage: signal reconstruction, next code prediction	-	Yes
FEMBA [65]	2D CNN, Mamba with forward and backward branches, CNN Decoder	Masked patch reconstruction	745M	No
FoME [44]	Temporal Transformer, Spatial Transformer	Masked patch reconstruction	-	Yes
GEFM [66]	GNN, wav2vec CNN, Transformer	Contrastive learning	-	No
LaBraM [52]	1st stage: Temporal CNN, VQ Codebook, Transformer; 2nd stage: Temporal CNN, Transformer	1st stage: Reconstruction (FFT amplitude and phase), Codebook loss; 2nd stage: Masked code prediction	5.8M-369M	Yes
LCM [67]	CNN, Transformer	Masked patch reconstruction	-	No
LEAD [54]	Temporal Transformer and Spatial Transformer	Contrastive learning	33.9M	Yes
M4CEA [68]	Temporal CNN, Transformer	Masked reconstruction (FFT amplitude and phase)	5.84M	Yes
MENDR [47]	GAT, 2D CNN, Transformer	Wavelet reconstruction	1.8M-6.5M	No
Neuro-GPT [45]	Temporal CNN, Transformer	Autoregressive next token prediction	78.5M	Yes
NeuroLM [69]	1st stage: Temporal CNN, Transformer, VQ Codebook; 2nd stage: Pre-trained LLM	1st stage: Reconstruction (signal and FFT amplitude), Codebook loss 2nd stage: Autoregressive next code prediction	254M-1.7B	Yes
SS-EMERGE [56]	Temporal GAT and Feature GAT, CNN	Contrastive learning	-	Yes

TABLE 2
Summary of Foundation Models trained on EEG data including scalp EEG and intracranial EEG (iEEG).

Mamba

Three of the EEG foundation models include Mamba blocks as the backbone of their architecture. The FEMBA [65] encoder consists of bidirectional Mamba blocks, each containing separate forward and backward branches before aggregation. The extension of Mamba, Mamba-2, has also been used in EEGMamba [60], with alternating forward and backward blocks. EEGM2 [42] comprises three separate Mamba-2 modules in an architecture similar to a U-Net: the Encoder progressively downsamples the input in the temporal resolution while increasing feature dimensionality,

the Decoder mirrors the Encoder by restoring temporal resolution, and the Mediator acts as the bottleneck between the two. Skip connections also allow local features from the Encoder to flow directly to the Decoder.

4.4 SSL Objectives

EEG foundation models are pre-trained using various SSL strategies including MAE, autoregressive and contrastive learning.

Many models use masking at some stage of the pre-training pipeline; this could be applied directly to input

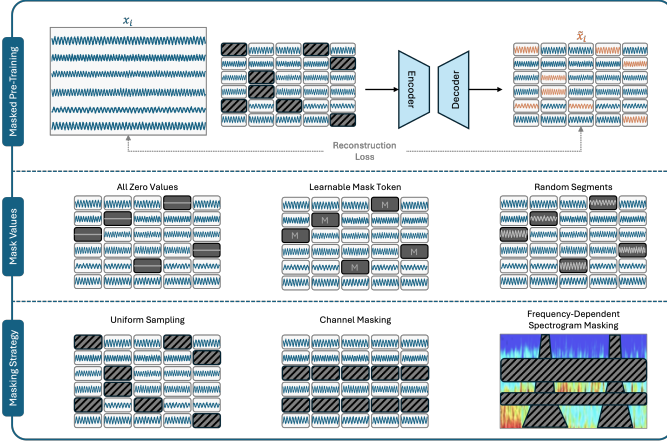


Fig. 5. Masking in biosignal foundation models. The masked pre-training paradigm (top), choices of mask values (middle), and strategies for selecting masked sections (bottom).

patches [59] or to latent representations like tokens [46], [65]. The mask itself could consist of all zero values [22], [65], a learnable mask token [44], [53], [55], or a random segment from the same sample [23]. A random portion of inputs is selected to be masked; this could be through uniform random sampling across dimensions [52], or masking out randomly selected channels [22]. Alternatives to uniform sampling include the frequency-dependent strategy proposed by BrainBERT [23] in which superlet spectrograms are masked in fewer, longer sections at lower frequencies. M4CEA [68] presents a unique approach by computing the mask probability for each patch using multiscale fluctuation-based dispersion entropy. An illustration of each masking strategy is presented in Fig. 5.

First stage tokenizers are all pre-trained to reconstruct their input in some way. This could mean reproducing the signal itself [43], [63], [69], frequency features such as FFT [52], [63], [69] and Discrete Prolate Spheroidal Sequences (DPSS) [53], or statistical features such as mean and standard deviation [43]. For all models involving VQ, the training objective also includes the following loss terms for training the codebook:

$$\mathcal{L}_{codebook} = \sum_i ||sg(h_i) - v_{z_i}||^2 \quad (13)$$

$$\mathcal{L}_{commitment} = \sum_i ||h_i - sg(v_{z_i})||^2 \quad (14)$$

$$\mathcal{L}_{VQ} = \mathcal{L}_{codebook} + \mathcal{L}_{commitment} \quad (15)$$

where h_i is the encoder output, v_{z_i} is the value of the nearest code, and $sg(\cdot)$ denotes the stop-gradient operator. In CodeBrain [63] the commitment loss term is ignored, and stop-gradient is only applied to the encoder parameters. The second stage model is then trained to predict the correct index in the codebook for each input, typically with cross-entropy loss.

Autoregressive training is employed by two of the GPT-style models: BrainGPT [50] and Neuro-GPT [45]. It is also used in the second stage of training EEGTrans [43] and NeuroLM [69] for next code prediction.

The SimCLR framework of contrastive learning is another popular pre-training strategy, although the selection of positive/negative pairs can vary. For example, LEAD [54] defines pairs on a sample or subject level. In SS-EMERGE [56], positive pairs may be any samples recorded with the same stimulus. BENDR [49] and GEFM [66] use a contrastive loss between the output of the wav2vec b , and the final output of the Transformer z : for position t , z_t forms a positive pair with b_t , and negative pairs include randomly selected distractors b_i from the same sequence.

4.5 Evaluation Methods

EEG foundation models are evaluated on their performance in downstream tasks after some adaptation process. The most common tasks are classification (e.g. sleep stage, seizure detection, emotion recognition) and regression (e.g. vigilance estimation), although some models including FoME [44] are also adapted for forecasting and imputation. It is most common to fine-tune the model for a single task, repeating the process for each benchmark dataset. However, some models including ALFEE [58], BrainGPT [50] and NeuroLM [69] use multi-task fine-tuning for all downstream tasks at once.

The adaptation process typically involves adding a simple classification head consisting of an FC layer or a lightweight MLP. The model is then either fully fine-tuned, or linear probing is applied; many publications choose to evaluate both options. In some exceptional cases, the classification head is more complex. ALFEE [58] trains a cross-attention layer where queries are taken from a task-specific classification token (CLS) dictionary. BrainGPT [50] uses a task-shared electrode graph network followed by task-specific heads. FEMBA [65] explores different prediction head options, comparing the performance of a linear classifier versus a Mamba classifier.

In addition to fine-tuning and linear probing, earlier publications such as BENDR [49] and BIOT [22] also train their models from scratch on downstream data, although this practice has since fallen out of fashion. EEGTrans [43] is evaluated on its ability to generate realistic EEG signals: their pre-trained model generates a synthetic training dataset for an EEGNet [74], which is then used to classify real unseen data.

5 ECG FOUNDATION MODELS

5.1 ECG Data and Features

An ECG trace shows a sequence of cardiac cycles, each containing components that reflect different physiological events that occur during a single heartbeat (including the P wave, QRS complex, and T wave).

The amplitudes and relative timings of these components can be highly informative in diagnosing cardiac conditions. [75] takes advantage of this by manually extracting features based on the RR interval (the time between R points in successive beats). Specifically, they calculate dRR , the difference between consecutive RR intervals, as this can be used to detect irregular heart rhythms such as atrial fibrillation (AFIB). However, the majority of foundation models choose to use raw ECG signals as input, after minor

preprocessing with resampling and z-score normalisation, for a more generalised approach.

The most widely used ECG datasets (see Table 8) contain 12-lead clinical recordings with diagnostic labels for abnormal beats, rhythms, and other cardiac conditions. The signals are represented as multi-channel time series data, with channels corresponding to leads. However, ECG leads are not independent in the same way as EEG electrodes: they all record electrical signals from the same underlying cardiac activity. As a result, several ECG models forgo spatial embeddings in favour of modelling the inter-channel relationships implicitly with convolutions. Alternatively, [21], [75] and Wearable-Echo-FM [76] all propose single-lead models.

5.2 Model Architecture

The most popular ECG foundation model architecture consists of a CNN encoder, followed by one or more Transformers. ECG-FM [77], for example, uses a model inspired by wav2vec 2.0 with convolutions, 2 VQ codebooks and a Transformer encoder.

Approaches that do not adhere to this design include [83] who choose not to include convolutions, and instead input patched ECG directly into a Transformer encoder. [21] explore three separate architectures (CNN, ResNet and Transformer) as unimodal models for ECG or PPG signals. [75] present a unique pipeline which creates a text corpus from encoded ECG features: *dRR* intervals are binned by length, and each bin is mapped to a single character. The characters are tokenized with Byte Pair Encoding (BPE), which operates on a sub-word level, and the resulting tokens are passed to the Transformer backbone.

Table 3 provides a summary of the ECG foundation model architectures, pre-training objectives and model sizes.

5.3 Leveraging Textual Reports

Hospital ECG datasets can include written reports from clinical experts, and it may be beneficial to utilise this information along with the corresponding signals. Wearable-Echo-FM [76] pre-trains a RoBERTa [84] LLM to encode textual reports, and applies 1D convolutions to encode raw ECG. The text and ECG encodings are then aligned with CLIP-style contrastive learning. ESI [80] adopts a similar strategy but with a pre-trained BioLinkBERT [85] LLM and, in addition to the alignment target, the paired ECG-text encodings are also passed to a shared decoder that learns to reconstruct the text.

KED [82] also uses clinical reports, however, the original diagnostic labels are augmented using LLMs to add necessary domain knowledge and explanations of medical terminology. A CNN and pre-trained BioClinicalBERT [86] encode ECG and text respectively. The core novelty of KED lies in its *Label Query Network* which applies cross-attention with encoded text labels as queries. To promote alignment in the shared latent space, ECG/text encodings are randomly selected as keys/values during pre-training.

5.4 Pre-training Objectives

ECG foundation models are most commonly trained using contrastive learning, including the approaches that use CLIP

to align ECG with written reports. Contrastive losses are also used without the text modality, such as in [21] where positive pairs are augmented views of the same sample, or samples collected from the same subject. Pairs may also be defined based on temporal adjacency as in ECG-FM [77] where consecutive patches are considered a positive pair. ECGFM [78] uses contrastive learning to predict future signal values by maximising similarity to the true values while minimising similarity with a set of randomly selected distractors.

In other cases, an MAE strategy is used to reconstruct ECG signal patches [83] or tokens [75]. In HuBERT-ECG [81], k-means clustering is applied to Mel features extracted from the raw signal, and the foundation model learns to predict cluster labels from masked signal patches.

An exception to these contrastive and MAE approaches is ECGFounder [79]. In this framework, positive-unlabelled (PU) learning is used in a supervised fashion for multi-label classification with partially labelled data. Clinical ECG data can have an extensive set of possible diagnostic labels, and while each sample may include some positive labels for certain conditions, the remaining unlabelled classes are not necessarily negative, and these are weighted accordingly to compute the loss.

5.5 Evaluation Methods

All ECG foundation models were evaluated with full fine-tuning and/or linear probing, with simple MLP heads for label prediction. Zero-shot evaluation is also applied in ESI [80] to assess the pre-trained model’s ability to retrieve the corresponding textual description for each ECG sample.

Downstream tasks usually involve multi-label ECG classification for a range of health targets, and can also include prediction of demographic labels such as subject age and sex [21], [79]. To assess cross-modal generalisation capability, ECGFounder [79] is further evaluated by fine-tuning on a PPG dataset for AFIB classification.

Most approaches perform classification on each recording; however, diagnostic labels for ECG can apply at different temporal scales. For example, left ventricular hypertrophy (LVH) is a chronic condition, meaning this label would apply throughout entire recordings for that subject. In contrast, a label of premature ventricular contraction (PVC), would only apply to the specific heartbeat. For this reason, ECGFM [78] performs beat-level or recording-level classification, depending on the task. [75] use an interval-based evaluation scheme in which AFIB/non-AFIB classification is applied to the ECG with a sliding window. Each heartbeat is then assigned a class label prediction by majority vote across all the windows containing that beat.

6 OTHER UNIMODAL BIOSIGNAL FOUNDATION MODELS

In this section, we review the four remaining unimodal biosignal foundation models found in our literature search. These include three for PPG and one for MEG signals. A summary of the models is provided in Table 4.

Model	Architecture	Pre-training Objective	Model Size	Open source
Abbaspourazad et al. [21]	CNN or ResNet or Transformer	Contrastive learning	2.5M-7.2M	No
BIOT [22]	FC Tokenizer, Transformer	Masked embedding reconstruction	3.2M	Yes
ECG-FM [77]	CNN, 2 VQ Codebooks, Transformer	Contrastive learning, Codebook diversity loss	90.9M	Yes
ECGFM [78]	CNN, Transformer	Contrastive Learning	19M	No
ECGFounder [79]	CNN, Group Convolutions and Channel-wise Attention	PU Learning	76.3M	Yes
ESI [80]	Pre-trained BERT-style LLM, ECG CNN, Cross-Modality Decoder	Contrastive learning	26.8-85.6M	Yes
HuBERT-ECG [81]	CNN, Transformer	Cluster prediction	30M-188M	Yes
KED [82]	ECG CNN, Pre-trained BERT-style LLM, Label Query Network Cross-Attention	Contrastive learning, Label prediction	164M	Yes
Moody et al. [83]	Transformer Encoder, Transformer Decoder	Masked signal reconstruction	92.7M	No
Tudjarski et al. [75]	BPE Tokenizer, Transformer	Masked token reconstruction	-	No
Wearable-Echo-FM [76]	1st stage: BERT-style LLM, 2nd stage: ECG CNN	1st stage: Masked token reconstruction, 2nd stage: Contrastive learning	-	No

TABLE 3
Summary of Foundation Models trained on ECG data.

Model	Modality	Architecture	SSL Objective	Model Size	Open source
Abbaspourazad et al. [21]	PPG or ECG	CNN or ResNet or Transformer	Contrastive learning	2.5M-7.2M	No
Csaky et al. [87]	MEG	Wavenet or Transformer	Autoregressive next token prediction	-	Yes
GPT-PPG [88]	PPG	Transformer	Autoregressive Logit-Laplace Distribution prediction	19M-1B	No
Pulse-PPG [89]	PPG	1st stage: CNN, Cross-Attention, 2nd stage: ResNet	1st stage: Masked signal reconstruction, 2nd stage: Contrastive learning	28.5M	Yes

TABLE 4
Summary of Foundation Models trained on unimodal biosignal data besides EEG and ECG.

6.1 Foundation Models for PPG

PPG signals have several unique characteristics to consider when developing a model. The signals are pulsative, reflecting periodic changes in blood volume corresponding to cardiac cycles, which means that most informative features occur at relatively low frequencies.

Data may be collected in a hospital or laboratory setting, but can also be acquired in the field with wearable devices such as smartwatches - although this exacerbates PPG's low SNR. Pulse-PPG [89] chooses to train on field data to improve the model's robustness to diverse noise sources, for better performance in real-world settings. They exploit short recurring patterns in PPG waveforms, known as "motifs", to organise the latent space in a semantically meaningful way. In the first stage of pre-training, a motif-based distance function is learned with MAE: masked portions of the signal are aligned with a bank of learnable motif embeddings using diluted convolutions and cross-attention. The learned distance function is then used to train the second stage ResNet encoder with contrastive learning: positive and negative pairs are determined by the distances between raw signals, and the contrastive loss function uses the distances between signal encodings.

[21] proposed various foundation models for ECG (reviewed in Section 5) and within the same paper they apply these architectures to PPG data with minor modifications. Due to the characteristics of PPG waveforms, the signals

are resampled to a lower frequency than that of ECG. PPG samples also exhibit high intra-subject variance, so the augmentations for SimCLR were applied with a lower probability. In addition, PPG augmentations included channel swap, which was not applicable for their single-lead ECG data. Spatial embeddings were not incorporated with multi-channel PPG since, like ECG, the underlying physiological activity for each channel is the same regardless of sensor location.

The GPT-PPG [88] model is composed of a stack of Transformer decoder blocks, which is pre-trained autoregressively. It was found that pre-training with an MSE loss on signal patches would cause the model to converge to a local minimum where the mean value was always predicted. The authors argue that this is because, once normalised, clean PPG signals are roughly symmetric about the mean. To address this issue, the model is instead pre-trained to predict the parameters of the logit-Laplace distribution.

Downstream tasks for evaluating PPG models include estimation of vital signs like heart rate and blood pressure, or predicting demographic information such as age and BMI.

6.2 Foundation Model for MEG

MEG measures the collective activity of synchronously firing neurons in the cortex. Unlike EEG, which records electric potentials through electrodes, MEG uses sensitive

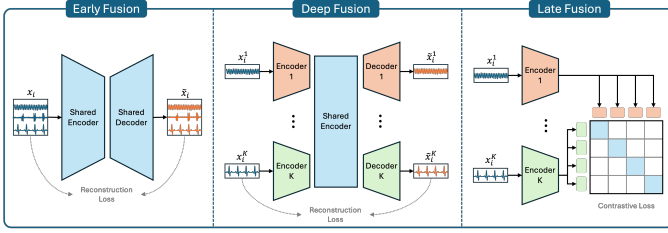


Fig. 6. Categories of multimodal model: Each illustration gives a simplified view of how a sample x_i , including data from K modalities, is processed in early, deep and late fusion approaches.

magnetometers to measure the associated magnetic fields. The advantage of this system is that magnetic fields are largely unaffected by the skull, leading to a higher SNR than for scalp EEG. Although iEEG also avoids signal disruption from the skull, MEG has the added benefit of being a non-invasive procedure. However, the recording process requires expensive equipment and must be conducted in a magnetically shielded environment, resulting in a scarcity of large-scale datasets.

[87] explores two architectures for modelling MEG: a Wavenet [90] and a GPT-style Transformer. In both cases, an initial "tokenization" step is performed in which a μ -law companding transform is applied and the data are quantised to 8-bits, and the model is always trained autoregressively to predict the next token. The Wavenet is applied independently to each channel, and an optional linear layer can be added to mix information across channels. For the GPT-style model, GPT2 blocks are applied to each channel in parallel, and spatial information is incorporated as a learnable embedding. Similar embeddings are also included to encode subject and task label information.

7 MULTIMODAL BIOSIGNAL FOUNDATION MODELS

From our literature search we found a total of six foundation models trained on more than one modality of biosignal, including EEG, EMG, EOG, ECG, MEG and respiration. It should be noted that every model used polysomnography (PSG) data during pre-training. For many PSG datasets, multiple physiological signals are recorded simultaneously, thereby facilitating alignment between modalities.

Each of the models approaches the problem of mixing modalities differently, with various architectures and training objectives, and can therefore vary considerably in model complexity and size as summarised in Table 5. The main distinction between these approaches is the point at which modality fusion occurs in the pipeline. In this section, we categorise the models into early fusion, deep fusion, and late fusion (illustrated in Fig. 6).

7.1 Early Fusion

BrainOmni [93] is a two-stage foundation model trained on EEG and MEG signals. Unlike the other multimodal models, the majority of its pre-training datasets are from a single modality and therefore have no natural "ground truth" alignment between EEG and MEG samples. Modality

information is incorporated using a type embedding, similar to spatial embeddings used in EEG. This is a learnable parameter that encodes information about the sensor type of each channel (EEG, MEG gradient or MEG amplitude). The type embedding is combined with a spatial embedding, based on the 3D coordinates of the sensor, to form a sensor embedding which is later used in the backbone of the tokenizer.

The first step of the BrainOmni tokenizer is a SEANet [97] that uses convolutions to encode temporal features of the raw signal. A cross-attention layer uses learnable queries to attend to keys formed by combining the sensor embedding and temporal encoding, with values from the temporal encoding. Then, an RVQ codebook is used to find the nearest codes for the latent representations. The tokenizer's decoder consists of another cross-attention layer, with queries taken from the sensor embedding, and another SEANet to reconstruct the signal. The tokenizer is pre-trained to reconstruct time and frequency features, and the codebook is updated using exponential moving average (EMA).

The second stage model of BrainOmni consists of a stack of Criss-Cross Transformer blocks, as introduced in CBraMod [48]. Input signals are tokenized with the pre-trained tokenizer, and the model learns to predict the masked codes. We refer to this approach as early fusion because input data from both modalities are essentially treated as a combined "EMEG" signal, and are processed by shared modules throughout the pipeline.

7.2 Deep Fusion

For deep fusion, we consider models that include modality-specific modules but which combine features in the intermediate layers.

The foundation model proposed by [91] has a simple but intuitive architecture composed of modality-specific tokenizers and decoders, with fusion occurring in an intermediate encoder. Each modality is tokenized by a separate 1D CNN, after which sinusoidal temporal embeddings and learnable modality embeddings are added. All tokens are encoded by a shared Transformer, and from this latent representation *every* decoder attempts to reconstruct the signals for its own specific modality. The model is pre-trained with an MAE objective, and also randomly drops entire modalities from input in order to encourage cross-modal reconstruction.

PhysioOmni [95] is a two-stage model that extends the LaBraM [52] architecture to multiple modalities. The tokenizer is composed of several modality-specific Transformer autoencoders, each of which uses its own VQ codebook in addition to one codebook shared across all modalities. In order for all modalities to use the same codebook, each of their encodings are temporally aligned with EEG features as an anchor. For unimodal reconstruction, the specific codes and shared codes are combined and passed to the modality's decoder. For cross-modal reconstruction, the specific codes are combined with temporally aligned EEG codes. The second stage trains the LaBraM encoders with MAE to predict both the specific and the shared codebook indices.

The architecture of the bioFAME [92] model can also be considered as deep fusion, although it does not explicitly

Model	Modalities	Architecture	SSL Objective	Model Size	Open source
Fang et al. [91]	EEG, EMG, EOG, ECG	Separate CNN Tokenizers, Shared Transformer Encoder, Separate Transformer Decoders	Masked signal reconstruction	-	No
bioFAME [92]	EEG, EMG, EOG	Modality-Independent Shared Encoder, Shared Transformer Autoencoder	Masked signal reconstruction	243K	Yes
BrainOmni [93]	EEG, MEG	1st stage: Tokenizer with SEAnet, Cross-Attention, RVQ Codebook; 2nd stage: Criss-Cross Transformer	1st stage: Reconstruction (signal, FFT amplitude and phase), Codebook loss; 2nd stage: Masked code prediction	8.4M-33M	No
Brant-X [94]	EEG, EMG, EOG, ECG	Separate Encoders: Temporal and Frequency CNNs, Transformer	Contrastive learning	1B	No
PhysioOmni [95]	EEG, EMG, EOG, ECG	1st stage: Separate LaBraM Tokenizers with Separate and Shared VQ Codebooks; 2nd stage: LaBraM	1st stage: Signal reconstruction, Codebook loss; 2nd stage: Masked code prediction (Private and Shared)	-	Yes
SleepFM [96]	EEG, EMG, EOG, ECG, Respiration	Separate CNN Encoders	Contrastive learning	-	Yes

TABLE 5
Summary of Foundation Models trained on multimodal biosignal data.

contain modality-specific modules. A shared frequency-aware encoder processes each channel (and by extension each modality) independently before channel mixing occurs in a shared encoder Transformer. A unique contribution in the architecture is the multi-head filter layer of the frequency-aware encoder: DFT is applied to the signal, followed by a bank of frequency filters, and IDFT inverts the features back to the temporal space. For decoding, the authors also explore the use of a single shared Transformer versus modality-specific Transformers.

7.3 Late Fusion (Contrastive Learning)

Late fusion approaches are models with no shared components between modalities, instead using contrastive learning to align latent representations of paired signals generated by separate encoders.

Brant-X [94] creates a framework to align EEG with simultaneously collected "EXG" signals (an placeholder term for EOG, ECG or EMG). The authors leverage their pre-trained EEG foundation model Brant-2 [62] to act as their EEG encoder. The architecture of the EXG encoder is similar to Brant-2, where CNNs are used to capture temporal and frequency information before patches are fed into a Transformer backbone. The encoded EEG and EXG signals are aligned on a patch and sequence level with the CLIP method using a similarity matrix. To improve robustness to different sampling rates, the input EXG data are also augmented with upsampled and downsampled patches.

SleepFM [96] is composed of three separate CNN encoders for ECG, Respiratory and "BAS" (ECG, EOG and EMG) signals. Within each encoder, channels are mixed with dilated and multi-channel 1D convolutions, without any additional embeddings relating to sensor types. Since raw signals are processed directly without patching, contrastive learning is applied at the sequence level between paired samples. Using a leave-one-out system, they define a positive pair with an embedding x_i^k from modality k as the average of the other two modalities $\bar{x}_i^{\neq k}$, while the negatives are the average embeddings for other sequences in the batch $x_j^{\neq k}$. This method was shown to improve

downstream performance compared to standard pairwise contrastive learning. SleepFM could also be considered an early fusion model in the sense that "BAS" signals are inherently multimodal and are technically fused within their shared encoder.

7.4 Evaluation Methods

The multimodal foundation models have been evaluated in a similar manner to the unimodal models: by assessing downstream performance after fine-tuning or linear probing. In most cases, a simple linear prediction head is added after the foundation model encoder. However, PhysioOmni [95] proposes a more complex pipeline where the features from pre-trained modality-specific encoders are mapped to a shared space, fused with an additional Transformer, and then aligned with a set of learnable prototypes (one for each class label). During fine-tuning, this prototype alignment loss is used in addition to an overall classification loss, and modality-specific losses to preserve independence in the encoders.

Besides downstream classification, the SleepFM [96] framework is evaluated for its cross-modal retrieval capability. Given a sample's embedding in one modality, its true pair in the other modalities is expected to rank as the most similar out of the test set. Median rank of the true pair, and how often it appeared in the top 10, were used as metrics.

Qualitative analysis is also performed through the use of visualisations of latent features. Brant-X [94] uses similarity matrices to show the learned association between EEG and EXG patch representations from the same sequence. Attention rollout, grouped by modality, is presented in [91] and bioFAME [92] to show connections between modalities after pre-training. BrainOmni [93] displays attention weights as topographic maps to show the brain regions associated with each query and head. Topomaps of real and reconstructed signals are also used for a visual comparison.

8 DESIGN CONSIDERATIONS

The development of biosignal foundation models involves several key choices that strongly influence efficiency and

generalisation. These include tokenization strategy, model architecture and pre-training objectives. Sections 4 to 7 have summarised the current state of the field; in this section we provide practical guidance on when and why to select specific designs based on evidence from the literature.

8.1 Tokenization Strategy Selection

The tokenization process determines how continuous signals are discretised for model input, balancing temporal resolution against computational efficiency. Point-based tokenization retains every time step (or with minimal down-sampling), thus preserving high frequency dynamics. This approach is suited to signal analysis where fine-grained temporal resolution is critical. For example, EEGM2 [42] applied multi-scale CNNs directly to raw EEG to capture detailed temporal dependencies. In contrast, patch-based tokenization segments the signal into fixed-length patches, reducing sequence length and computational cost while maintaining contextual information. This has become the dominant strategy, used by 32 of the 50 surveyed biosignal foundation models. Overlapping patches (e.g. BIOT [22]) offer smoother reconstruction, but increase redundancy, whereas non-overlapping patches provide greater efficiency at the cost of boundary precision. CNN-based patch extraction, such as in LaBraM [52], offers a practical balance between efficiency and feature richness. Tokenisation can also be guided by domain knowledge of the specific modality, such as aligning to heartbeats in ECG or PPG traces. An example is the motif-based segmentation used in Pulse-PPG, which preserves the intrinsic physiological structure.

8.2 Architecture Selection

Transformer architectures are found to be the most widely adopted due to their interpretability and the modelling capabilities of self-attention, however their quadratic complexity limits scalability for long recordings. State-space models such as Mamba achieve linear complexity with respect to sample length, and are therefore better suited to longer continuous data while maintaining competitive representational abilities. For these reasons, Mamba architectures are advantageous under constrained computational budgets, and Transformers should be preferred when interpretability is a priority. The choice of structure in a Transformer based model should also be carefully considered. BERT-style models learn bidirectional representations that are well suited to classification, GPT-style models are effective for autoregressive generation and forecasting, and full encoder-decoder Transformers are the best choice for sequence-to-sequence or translation tasks. For multi-channel biosignals such as EEG, spatial relationships require further handling. Fixed channel ordering offers simplicity but restricts generalisation to a specific electrode montage. Learnable spatial embeddings are the most common approach, as these enable adaptation across different electrode layouts. Graph-based representations, as in GEFM [66] and MENDR [47] explicitly encode the inter-electrode geometry to provide superior spatial modelling. However this introduces additional complexity and requires annotations for the relative positions of electrodes.

8.3 Pre-training Objective Selection

The nature of the available pre-training data and the desired downstream capabilities will influence the appropriate choice of SSL objective. When paired modalities are present, such as EEG-EMG or ECG-text, contrastive alignment using CLIP is preferred [80], [94], [96]. When reconstruction interpretability is a priority, MAE objectives can provide direct access to the reconstructed signal or features [22], [46], [48]. Autoregressive pre-training is most appropriate if the intended downstream tasks include forecasting. Finally, contrastive learning with a SimCLR objective is a widely used method for a flexible approach [49], [54].

8.4 Key Insights from Literature

Across the surveyed works, several clear trends emerge. Patch-based tokenization is dominant due to its favourable balance between resolution and efficiency. Spatial embeddings can improve performance over channel-agnostic designs by capturing electrode information to better model inter-channel relationships. Multi-scale feature extraction improves robustness and adaptability across tasks compared to single-scale. Two-stage VQ pre-training enables discrete reasoning at the cost of additional complexity.

In practice, an initial baseline for most applications could consist of a patch-based tokenization, combined with a Transformer encoder and an MAE objective. This reflects the most common and well established approach from the current literature. For limited computational resources, a Mamba backbone paired with contrastive learning offers efficient scaling to long sequences. When working with multimodal biosignal data, an effective solution is to train separate encoders with contrastive alignment to preserve modality integrity. Lastly, when developing models for clinical deployment, attention-based models are preferred for their interpretability.

9 CHALLENGES AND FUTURE DIRECTIONS

9.1 Fair Comparisons in Benchmarking

The capabilities of biosignal foundation models have been evaluated in a myriad of ways, with adaptation strategies and downstream tasks varying from model to model, making effective and reliable comparisons difficult. Although the increased availability of biosignal corpora has been advantageous for pre-training these large models, it has also caused a divergence in which datasets are selected for downstream evaluation. It is standard practice to compare performance against baseline models; however this is only possible in cases where code and pre-trained weights for the baselines are made available. Data leakage can contaminate results if downstream data were used during any model's pre-training, and results may also vary depending on specific fine-tuning hyperparameters or other implementation details.

Standardised benchmark sets for EEG have been proposed, although these have yet to be widely adopted. These can be paradigm-specific, such as LibEER [98] which focuses on emotion recognition datasets, and SzCORE [99] which evaluates seizure detection. Other EEG benchmarks have recently been proposed to assess model generalisability across

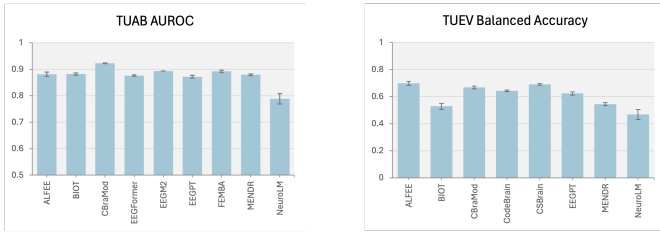


Fig. 7. Comparison of EEG foundation model classification performance on TUAB (left) and TUEV (right).

a range of tasks, including ALBM [100] and AdaBrain-Bench [101]. ALBM provides a subject-independent evaluation protocol, in which the test set only includes unseen subjects. They include five BCI tasks with datasets specifically chosen to prevent spurious correlations (i.e. models fitting to task-discriminative artifacts in the data). AdaBrain-Bench proposes a selection of 13 EEG datasets, evaluated under subject-independent and subject-dependent settings. They also include few-shot transfer settings, where the amount of fine-tuning training data is reduced by varying degrees.

To provide some insight into the relative performance of the EEG foundation models, we compare results for the most commonly used downstream datasets. These are two subsets of TUEG: TUAB for normal/abnormal classification. The results presented in Fig. 9.1 show that CBraMod achieves the best classification for TUAB, outperforming larger models such as FEMBA and NeuroLM. ALFEE shows impressive performance on TUEV after multi-task fine-tuning, achieving better classification than models that had been specifically adapted to the single task. EEGPT also shows good generalisation capability as its performance after linear probing is competitive with others' fully fine-tuned models, even without including any TUAB or TUEV samples during pre-training.

The Physionet Computing in Cardiology Challenge 2021 [102] is an established benchmark set for ECG models. It contains several datasets of 12-lead recordings from China, USA and Europe including: Chapman-Shaoxing, CPSC and CPSC Extra, Georgia, Ningbo, PTB and PTB-XL. Over half of the ECG models reviewed in this survey were evaluated on PTB-XL, making it the most popular downstream dataset for ECG foundation models: we present the classification results in Table 6. However, even when using the same evaluation dataset, direct comparison of performance is still a challenge due to differences in the exact task being performed. PTB-XL includes 71 different labels covering Diagnostic, Form and Rhythm statements, but most models choose to use only a subset of these for multi-label classification. KED selects 46 labels from across the three categories, excluding those with less than 100 samples. ECGFounder and ECGFM include 30 of the Diagnostic classes, whereas BIOT and ESI use only Rhythm statements for arrhythmia detection. HuBERT-ECG is the only model to perform classification with all 71 labels, while also presenting AUROC scores for the 46-label and 30-label problems.

Of the three PPG models, GPT-PPG and Pulse-PPG are both evaluated on the two datasets WESAD [103] and DaLiA [104]. However, the problem again arises where

Model	71-label	46-label	30-label	Arrhythmia
BIOT	-	-	-	0.766
ECGFM	-	-	0.917	-
ECGFounder	-	-	0.924	-
ESI	-	-	-	0.938
HuBERT-ECG	0.903	0.911	0.921	-
KED	-	0.858	-	-

TABLE 6

Classification performance (AUROC) of ECG foundation models on PTB-XL and its class subsets. Values correspond to the best performing variant of the model, as reported by the authors.

the same data are being used for different downstream tasks. The one common task between the two is heart rate estimation on the DaLiA dataset, for which GPT-PPG and Pulse-PPG achieve a mean absolute error of 4.77 and 3.71 respectively.

It is not currently possible to directly compare the multi-modal models in this review as they do not share common evaluation datasets. Neither do they benchmark against each other as baselines, instead choosing to compare results with earlier unimodal models or supervised multimodal methods.

Following the direction of other foundation models in language and image processing, we should expect to see the introduction and adoption of comprehensive biosignal benchmark sets. It is vital to have an effective evaluation of models' relative strengths and weaknesses over a range of tasks including classification/regression, but also generative applications such as data denoising, corrupted signal repair, and imputation. This will lead to more directed approaches in model development to address known limitations of previous models.

9.2 Explainability and Interpretability

Interpretability is a central challenge for biosignal foundation models. Unlike in domains such as vision or language, raw biosignals are already abstract representations of underlying physiological processes. It is therefore especially important to make model decisions understandable, not only to other researchers but also to end users, such as clinicians, who need trust and transparency to apply these systems in high-stakes contexts. Clear explanations of model predictions are required to validate whether the reasoning is physiologically plausible or if it is fitted to spurious correlations.

Several approaches have been explored to examine how data are processed and used within biosignal models. For example, many choose to perform ablation studies that reveal the value of specific architectural choices or pre-training objectives by systematically removing them, or replacing them with simpler alternatives, before comparing downstream performance. Attention maps have been used to visualise the learned connections, particularly between modalities in multimodal models. Saliency maps could also be a useful method to suggest which temporal or spatial regions of a signal most influenced a decision. Perturbation-based tests represent another promising direction: by modifying the input signal (e.g. jittering, frequency band occlusion or channel dropout) and observing the impact on

output predictions, this can provide insight into what information the model truly relies on. These model-centric interpretability methods can help build a clearer picture of which aspects of architecture and pre-training are most impactful, and in what contexts.

Another source of opacity lies in pre-processing decisions. Practices such as bandpass filtering and resampling to a uniform frequency are widely adopted, but each risks discarding salient information. Bandpass limits are often selected based on convention or domain knowledge of relevant frequency bands, but the trade-off between artifact removal and information loss is not quantified. Similarly, resampling simplifies the training process for heterogeneous signals, but may downsample away important dynamics at higher frequencies. Future work could benefit from multi-scale and multi-resolution approaches to preserve more physiologically meaningful details.

For models pre-trained with MAE objectives, interpretability can also be improved by reporting reconstruction fidelity, including quantitative and qualitative results for transparency and clarity. Combined with zero-shot learning or linear probing, this can reveal whether better reconstruction then translates to more useful latent representations and improved downstream performance.

9.3 Data Availability and Underrepresented Modalities

Although the amount of publicly available biosignal data has grown considerably in recent years, their scale remains small relative to the massive corpora used to pre-train models in text and vision. The largest collections to date come from hospitals, limiting their data to recordings in controlled environments for medical applications. However, consumer-grade devices are gradually becoming more affordable, enabling collection of real-world field data. We can expect this shift to broaden the diversity of pre-training data and thus improve model generalisability to non-clinical tasks such as human-computer interaction.

Demographic biases in training data also remain a concern. Hospital datasets may overrepresent patients with certain common conditions, while non-clinical data are often collected in universities which can lead to skewed age distributions. Biosignal foundation models typically mitigate this by utilising multiple datasets that span different populations and recording conditions. However, diversity in ethnicity and nationality is also limited, with the most popular large datasets originating from only a handful of countries. The collection of demographically diverse data will be essential for training models that can generalise across populations in real-world use.

Progress in multimodal foundation models depends heavily on large-scale datasets with synchronously recorded signals. PSG studies have been a primary source, but broader coverage is needed across different paradigms for more general-purpose models.

Certain biosignal modalities remain underrepresented in foundation model research. EEG and ECG have received the most attention in the literature, reflecting the availability of large datasets. In contrast, we found no EMG foundation models, despite a number of published EMG-specific transformer architectures. Barriers to this could include a

lack of large-scale corpora, and the heterogeneity in sensor placements across datasets. The lack of a standardised sensor layout poses a serious issue for EMG compared to other modalities. Similarly, PPG has had limited exploration in just three foundation models. Given that PPG can be collected cost-effectively and at scale with consumer wearables, it holds considerable promise for training both unimodal and multimodal models.

9.4 Readiness for Real World Application

To move beyond proof-of-concept studies into real-world use, practical considerations such as efficiency, scalability and privacy will need to be addressed. A key challenge is ensuring that models can run inference in real-time on devices with limited computational resources. There are several existing strategies to address this without compromising performance, including model compression [89], [94], pruning [49], and knowledge distillation [22]. Efficient architectures and optimised inference pipelines are vital in making foundation models usable outside of research environments.

Transformer architectures currently dominate foundation models, but the complexity of the attention mechanism scales quadratically with sequence length. This issue is alleviated in most Transformer based biosignal models which use short samples segmented into patches. The disadvantage is that biosignals can be long, continuous recordings with long-range temporal dependencies that could be missed. A promising future direction is the increased use of SSM architectures such as Mamba, which promise linear-time complexity while preserving long-range dependencies.

Deployment also requires rigorous frameworks for privacy-preserving training and ethical validation. For example, federated learning would enable models to be trained on sensitive data without moving the raw data from the source institution. Clinical integration of any model will require careful validation under regulatory standards, such as the EU Medical Device Regulation or the FDA Software as a Medical Device framework, to ensure safety and reliability.

Inter-subject variability is particularly pronounced in biosignal data, causing a significant challenge in generalisability. Personalisation may be used to address this, using a lightweight calibration process to tune the model to the individual's physiology. Designing foundation models that can adapt quickly and reliably to new users without requiring extensive re-training will enable robust real-world adoption.

10 CONCLUSION

Foundation models, having become the dominant paradigm in Natural Language Processing (NLP) and computer vision, are now beginning to shape research in biosignal analysis. In this survey, we provide a comprehensive overview of the current state of the field. We introduced the major biosignal modalities and their defining characteristics that make them both promising and challenging for training large models. Reviewing these models, we examined the most popular data sources, input features, embeddings, architectures, pre-training objectives, and evaluation practices. We also discussed the different fusion strategies adopted by

each of the multimodal approaches from early fusion to late fusion.

To summarise our findings: a growing number of foundation models have been trained on EEG or ECG signals. PPG has recently begun to appear in unimodal settings, and EMG remains underrepresented despite its practical applications. Multimodal biosignal models are an especially promising direction, but progress in this area remains limited by the availability of large datasets containing synchronously recorded signals. Transformers are decidedly the most popular backbone architecture across all modalities, although a few Mamba based models have been proposed for EEG.

We identified several challenges and opportunities that we believe could shape the future trajectory of this field. We reviewed many different training and architectural choices in the literature, but comparing their relative efficacy will require fair benchmarking and reproducible evaluation. Interpretability must also be prioritised to understand and explain model decisions, which will be necessary to build trust before they can be used in clinical contexts. Moreover, real-world deployment will require fast inference in resource-constrained devices using noisy, heterogeneous, and often sensitive data. This presents an opportunity for advances in multi-resolution modelling, federated learning, and efficient architectures to improve readiness.

The development of biosignal foundation models is still in the relatively early stages compared to their counterparts in text and vision. However, they hold great promise if challenges in data diversity, interpretability and evaluation can be addressed. Following current momentum in this field, we may soon see robust, generalisable models with the potential to transform how biosignals are used in real-world applications.

11 ACKNOWLEDGMENTS

This work was supported by the EPSRC Turing AI Fellowship (Grant Ref: EP/Z534699/1): Generative Machine Learning Models for Data of Arbitrary Underlying Geometry (MAGAL).

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APPENDIX A

SUPPLEMENTARY MATERIAL:

COMMONLY USED EEG DATASETS

TABLE 7: Common EEG datasets and foundation models which use them for pre-training or downstream tasks. We include only datasets that have been used by at least two models.

Dataset (Paradigm)	Pre-training	Downstream
BCI IV-1 [105] (Motor Imagery)	EEGTrans, LaBraM, NeuroLM	BrainGPT
BCI IV-2a [105] (Motor Imagery)	-	BENDR, CSBrain, EEGPT, EEGTrans, LCM, Neuro-GPT
BCI IV-2b [105] (Motor Imagery)	EEGTrans	EEGPT, LCM
BCIC2020-3 [106] (Imagined Speech)	-	CBraMod, CSBrain, EEGMamba
bi2015a [107] (ERP)	ALFEE, LaBraM, NeuroLM	-
CHB-MIT [108] (Resting)	BIOT, FoME	BIOT, Brant-2, CBraMod, CodeBrain, CSBrain, EEGMamba
Citi et al. [109] (ERP)	-	BENDR, GEFM
EEGMat [110] (Workload)	-	ALFEE, BrainGPT, CBraMod, CodeBrain, CSBrain, NeuroLM
Emobrain [111] (Emotion Recognition)	ALFEE, LaBraM, NeuroLM	-
FACED [112] (Emotion Recognition)	BrainGPT	BrainGPT, CBraMod, CodeBrain, CSBrain, DIVER-0, EEGMamba
FNUSA [113] (Seizure Detection)	FoME	Brant-2, FoME
Grasp and Lift [114] (Motor Execution)	ALFEE, LaBraM, NeuroLM	-
High Gamma [115] (Motor Imagery)	EEGPT	EEGTrans
HMC [116] (Sleep Staging)	-	ALFEE, BrainGPT, Brant-2, CSBrain, NeuroLM
Inria P300 [117] (ERN)	ALFEE, LaBraM, NeuroLM	BENDR, EEGPT, GEFM
ISRUC [118] (Sleep Staging)	-	CBraMod, CodeBrain, CSBrain, EEGMamba
MAYO [113] (Seizure Detection)	FoME	Brant, FoME
MoBI [119] (Motor Imagery)	-	CEReBrO, LaBraM
Mumtaz [120] (Mental Disorder Diagnosis)	-	CBraMod, CSBrain
Neonate [121] (Neonatal Seizure Detection)	-	CEReBrO, EEGFormer
Physionet MI [122] (Motor Imagery)	ALFEE, EEGPT, FoME, LaBraM, LCM, NeuroLM	BENDR, Brant-2, CBraMod, CSBrain, DIVER-0, EEGMamba, GEFM
SEED [123] (Emotion Recognition)	ALFEE, BrainGPT, EEGPT, FoME, LaBraM, LCM, SS-EMERGE	ALFEE, Brant-2, CEReBrO, FoME, NeuroLM, SS-EMERGE
SEED-FRA [124] (Emotion Recognition)	ALFEE, BrainGPT, LaBraM, NeuroLM	-

Dataset (Paradigm)	Pre-training	Downstream
SEED-GER [124] (Emotion Recognition)	ALFEE, BrainGPT, LaBraM, NeuroLM	-
SEED-IV [125] (Emotion Recognition)	ALFEE, BrainGPT, FoME, LaBraM, NeuroLM, SS-EMERGE	BrainGPT, SS-EMERGE
SEED-V [126] (Emotion Recognition)	ALFEE, BrainGPT, NeuroLM	BrainGPT, CBraMod, CodeBrain, CSBrain, LaBraM
SEED-VIG [127] (Vigilance Estimation)	-	CBraMod, CSBrain
SHU-MI [128] (Motor Imagery)	-	CBraMod, CodeBrain, CSBrain
Siena Scalp [129] (Epilepsy)	ALFEE, EEGMamba, LaBraM, NeuroLM	Brant-2, CSBrain
Sleep-EDF [130] (Sleep Staging)	FoME	BENDR, BrainGPT, Brant-2, EEGPT, FoME
SPIS Resting [131] (Resting)	ALFEE, LaBraM, NeuroLM	-
STEW [132] (Workload)	EEGM2	BrainGPT, EEGM2
Trujillo et al. [133] (Resting)	ALFEE, LaBraM, NeuroLM	-
Trujillo [134] (Workload)	ALFEE, EEGMamba, LaBraM, NeuroLM	-
TUEG [41] (Clinical)	ALFEE, BENDR, Brant-2, CBraMod, CEReBrO, CodeBrain, CSBrain, DIVER-0, EEGFormer, EEGMamba, FEMBA, FoME, GEFM, MENDR, Neuro-GPT, NeuroLM	-
TUEG - TUAB [135] (Abnormal/Normal)	ALFEE BENDR, BioSerenity-E1, BIOT, Brant-2, CBraMod, CodeBrain, CSBrain, DIVER-0, EEGFormer, EEGM2, EEGMamba, FoME, GEFM, MENDR, Neuro-GPT, NeuroLM	ALFEE, BioSerenity-E1, BIOT, CBraMod, CEReBrO, CodeBrain, CSBrain, EEGFormer, EEGM2, EEGPT, FEMBA, MENDR, NeuroLM
TUEG - TUAR [136] (Artifacts)	ALFEE, BENDR, BioSerenity-E1, Brant-2, CBraMod, CEReBrO, CodeBrain, CSBrain, DIVER-0, EEGFormer, EEGMamba, FoME, GEFM, LaBraM, MENDR, Neuro-GPT, NeuroLM	EEGFormer, FEMBA
TUEG - TUEP [137] (Epilepsy)	ALFEE, BENDR, Brant-2, CBraMod, CEReBrO, CodeBrain, CSBrain, DIVER-0, EEGFormer, EEGMamba, FEMBA, FoME, GEFM, LaBraM, LEAD, MENDR, Neuro-GPT, NeuroLM	-
TUEG - TUEV [138] (Events)	ALFEE, BENDR, BioSerenity-E1, BIOT, Brant-2, CBraMod, CEReBrO, CodeBrain, CSBrain, DIVER-0, EEGFormer, EEGMamba, FEMBA, FoME, GEFM, MENDR, Neuro-GPT, NeuroLM	ALFEE, BIOT, CBraMod, CodeBrain, CSBrain, EEGPT, FoME, MENDR, NeuroLM
TUEG - TUSL [139] (Slowing Events)	ALFEE, BENDR, Brant-2, CBraMod, CEReBrO, CodeBrain, CSBrain, DIVER-0, EEGFormer, EEGMamba, FoME, GEFM, LaBraM, MENDR, Neuro-GPT, NeuroLM	ALFEE, CSBrain, EEGFormer, FEMBA, NeuroLM
TUEG - TUSZ [140] (Seizure Type)	ALFEE, BENDR, BioSerenity-E1, Brant-2, CBraMod, CEReBrO, CodeBrain, CSBrain, DIVER-0, EEGFormer, EEGMamba, FEMBA, FoME, GEFM, LaBraM, MENDR, Neuro-GPT, NeuroLM	BioSerenity-E1, EEGFormer

APPENDIX B

SUPPLEMENTARY MATERIAL: COMMONLY USED ECG DATASETS

Dataset	Pre-training	Downstream
Chapman-Shaoxing [141]	ECG-FM, ESI	ECGFM, HuBERT-ECG, KED
CODE [142]	-	ECGFounder, HuBERT-ECG
CPSC and CPSC Extra [143]	ECG-FM	ECGFM, HuBERT-ECG, KED
Georgia [144]	ECG-FM	ECGFM, HuBERT-ECG, KED
HeFei [145]	-	ECGFM, HuBERT-ECG
MIMIC-IV-ECG [146]	ECG-FM, ESI, KED, Moody et al.	ECG-FM, ECGFounder
MIT-BIH [147]	Tudjarski et al.	ECGFM, Tudjarski et al.
Ningbo [148]	ECG-FM	ECGFM, HuBERT-ECG
PTB [149]	-	ECGFM, HuBERT-ECG
PTB-XL [150]	BIOT, ECG-FM, ESI	ECGFM, ECGFounder, ESI, HuBERT-ECG, KED

TABLE 8

Common ECG datasets and the foundation models that use them for pre-training or downstream tasks. We include only datasets that have been used by at least two models.

APPENDIX C

SUPPLEMENTARY MATERIAL: OPEN-SOURCE BIOSIGNAL FOUNDATION MODELS

The following list provides links to code for all open-source biosignal foundation models reviewed in this survey.

Model	Source Code
BENDR	https://github.com/SPOClab-ca/BENDR
BIOT	https://github.com/ycq091044/BIOT
BrainBERT	https://github.com/czlwang/BrainBERT
CBraMod	https://github.com/wjq-learning/CBraMod
EEGM2	https://github.com/Emotiv/EEGM2-ICDM2025
EEGMamba	https://github.com/wjq-learning/EEGMamba
EEGPT	https://github.com/BINE022/EEGPT
EEGTrans	https://anonymous.4open.science/r/EEGTrans-Transformer-Driven-Generative-Models-for-EEG-Synthesis-0FD9
LaBraM	https://github.com/935963004/LaBraM
LEAD	https://github.com/DL4mHealth/LEAD
M4CEA	https://github.com/Evigouse/M4CEA_Project
Neuro-GPT	https://github.com/wenhui0206/NeuroGPT
NeuroLM	https://github.com/935963004/NeuroLM
SS-EMERGE	https://github.com/cahuja1992/ss-emerge
ECG-FM	https://github.com/bowang-lab/ECG-FM/
ECGFounder	https://github.com/PKUDigitalHealth/ECGFounder
ESI	https://github.com/comp-well-org/ESI
HuBERT-ECG	https://github.com/Edoar-do/HuBERT-ECG
KED	https://github.com/control-spiderman/ECGFM-KED
Csaky et al.	https://github.com/ricsinaruto/MEG-transfer-decoding
Pulse-PPG	https://github.com/maxxu05/pulseppg
bioFAME	https://github.com/apple/ml-famae
PhysioOmni	https://github.com/935963004/PhysioOmni
SleepFM	https://github.com/rthapa84/sleepfm-codebase